05-17 第2篇

阅读理解（说明文） Why Do We Have Fingerprints?

① If you look closely at the tip of your finger under a bright lamp, you will see small curved lines forming loops, arches or circles. These tiny patterns are your **fingerprints**. You have had them ever since you were a baby, and you will keep them for your whole life. They do not change as you grow, they do not come back differently after a small cut, and they are never exactly the same as anyone else's — not even an identical twin's. Why do humans have fingerprints?

② Scientists have studied this small puzzle for over a hundred years, and there are several interesting answers. The **first** and oldest idea is about **grip**. A flat, smooth finger might slide across a wet glass or a shiny book cover. A finger with tiny ridges creates more friction. When your fingertip touches a cup, the small lines press, bend slightly, and hold better. Holding chopsticks, a pencil or a glass of milk becomes easier with these lines than without.

③ A **second** useful role is about **touch**. Under each ridge on your fingertip, there are many tiny **nerve endings**. When your finger slides across a surface — a page, a fabric, a leaf — the ridges vibrate against the surface. These vibrations travel to the nerves and help the brain understand texture: rough, smooth, sticky, silky. Some experiments even suggest that the ridges make the fingertip about as sensitive as a blind reader's finger reading Braille.

④ A **third** idea is about **protecting skin** from damage. Our fingers meet hot, cold, rough or sharp things every day. The ridges may help spread the pressure over a wider area, the way the pattern on a bicycle tyre spreads weight. They may also help sweat flow in small rivers, which keeps the fingertip slightly damp but not slippery — a very useful balance when you want to turn a page quickly.

⑤ Fingerprints are also very useful outside the body. Because every person's ridges are **unique**, police officers have used fingerprints to solve crimes for more than 100 years. Some hospitals use fingerprints to identify newborn babies. Modern phones and laptops allow you to unlock them with just a gentle press of your finger. The quiet little lines that help you hold a pencil also help the world keep track of you.

⑥ Interestingly, we are still not 100% sure why our fingerprints form into so many different shapes. Scientists know that they are decided before birth, inside the mother's body, and that both **genes** and small physical movements in the womb play a part. But exactly why one baby has mostly loops and another mostly arches — that is still an open

question. Your own ten little patterns are a kind of personal signature, written on you before you were born.

根据短文内容，选择最佳答案。

1. Which of the following is TRUE about fingerprints?

A. They slowly change their shape every few years.
B. They are exactly the same in identical twins.
C. They do not change as we grow, and they come back after a small cut.
D. Babies are born without any fingerprints.

2. According to paragraph ②, how do fingerprint ridges help with grip?

A. They make the finger longer.
B. They create more friction, so the finger holds better.
C. They make the cups feel heavier.
D. They stop all sweat.

3. What does paragraph ③ say about the ridges and touch?

A. The ridges block the nerves.
B. Ridges vibrating against surfaces help nerves understand texture.
C. Ridges only feel cold surfaces.
D. Ridges make fingertips less sensitive.

4. Which of the following is NOT a modern use of fingerprints mentioned in the passage?

A. Police solving crimes.
B. Identifying newborn babies in some hospitals.
C. Unlocking phones and laptops.
D. Predicting tomorrow's weather.

5. What is the passage MAINLY about?

A. Why all identical twins are exactly the same.
B. How phones unlock without any password.
C. What fingerprints are, why we have them, and how they are used.
D. How to make a fake fingerprint at home.

答案与解析

1.
【答案】C

【题型】细节理解题
【定位】第①段 "They do not change as you grow, they do not come back differently after a small cut..."
【解析】指纹终生不变且伤后依原样恢复。A/B/D 均与原文相反。故选 C。

2.
【答案】B
【题型】细节理解题
【定位】第②段 "A finger with tiny ridges creates more friction."
【解析】脊线产生摩擦、便于抓握。A/C/D 均不符。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第③段 "the ridges vibrate against the surface. These vibrations travel to the nerves..."
【解析】脊线振动向神经传递信息。A/C/D 均与原文相反或无据。故选 B。

4.
【答案】D
【题型】细节理解题
【定位】第⑤段三项用途：破案、识别新生儿、解锁手机
【解析】预测天气未被提及，属无中生有。故选 D。

5.
【答案】C
【题型】主旨大意题
【定位】全文结构
【解析】文章介绍指纹是什么、我们为何有它们、及其应用。A/B/D 均非主旨。故选 C。

词汇注释（超纲词）

- **fingerprint** [n.] 指纹
- **ridge** [n.] 脊；垄
- **friction** [n.] 摩擦
- **Braille** [n.] 盲文